
DS-GA 3001 Course Project

Point Tracking in Egocentric Videos

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Abstract

Point tracking in egocentric videos presents significant challenges due to continuous camera motion, frequent occlusions, and dynamic object interactions. These factors demand spatially precise and robust feature representations. In this work, we explore the use of 2D foundation models—DINOv2 and CroCov2—for improving point correspondence across frames. We evaluate simple baselines that use frozen features from these models and introduce a lightweight delta network that learns displacement residuals between corresponding points. Our approach highlights the potential and limitations of adapting general-purpose foundation models for the fine-grained task of point tracking in egocentric video settings. Our code is available [here](#).

1 Introduction

Egocentric videos, captured from a first-person perspective, are gaining increasing attention in computer vision due to their relevance in applications such as augmented reality and robotics. As a result, there is the need to have robust computer vision algorithms for various scene understanding tasks in such videos. Tracking in egocentric videos is essential for understanding object interactions, maintaining situational awareness, and enabling real-time applications.

Egocentric video point tracking involves detecting and following points from the perspective of a moving camera, typically worn by a person or mounted on a robot. Unlike traditional tracking methods that rely on static or third-person viewpoints, egocentric tracking must contend with continuous motion, abrupt viewpoint shifts, severe occlusions, and dynamic interactions with the environment. Additionally, it requires long-term point association, ensuring consistent re-identification of points as they appear, disappear, and reappear over extended periods, potentially throughout a user's lifetime. These challenges make it a crucial area of research for applications in augmented reality, robotics, and assistive technologies. Effective tracking in egocentric videos demands models that can robustly integrate temporal information, depth cues, and motion patterns while adapting to rapid perspective changes, enabling advancements in hands-free navigation, activity recognition, and human-object interaction analysis.

2 Related Works

Video point tracking involves following a sparse set of points from an initial (query) frame to subsequent (target) frames, handling occlusions and appearance changes. Unlike optical flow, which computes pixel-wise correspondences between adjacent frames, point tracking focuses on maintaining long-term associations of specific points through occlusions and out-of-view intervals.

These persistent trajectories support downstream applications such as Structure-from-Motion and video editing.

The problem of point tracking in third-person videos is well-established, with extensive research leading to the development of robust and highly accurate tracking methods. Numerous algorithms have been proposed, including convolution-based trackers [1], transformer-based models [2], and probabilistic filter-based methods [3], each refining the ability to track points across challenging scenarios such as occlusions, illumination changes, and fast motion. These methods have been extensively evaluated on large-scale benchmarks such as TAP-Vid DAVIS[4], TAP-Vid Kinetics[5], and FlyingThings++[1], achieving impressive results that demonstrate their effectiveness in real-world applications. Additionally, third-person tracking benefits from relatively stable viewpoints, global scene context, and well-defined object appearances, which make tracking more reliable. However, despite these advances, transferring the same principles to egocentric video tracking remains challenging due to continuous viewpoint shifts, frequent occlusions, and dynamic interactions with objects, necessitating new solutions tailored to first-person perspectives.

The PIPs framework [1] treats each pixel as an independent 'particle' and iteratively refines its trajectory via multi-frame correlation volumes, effectively handling occlusions without explicit motion priors. CoTracker3 [2] uses a transformer-based network and semi-supervised labeling from existing architectures to annotate real videos, achieving state-of-the-art performance with 1,000x less synthetic data. DINO-Tracker [6] leverages self-supervised semantic priors from a frozen DINO backbone combined with test-time adaptation on a single video to achieve long-term dense tracking, fine-tuning DINO features and a lightweight tracker end-to-end. ProTracker [3] incorporates optical flow and semantic correspondences within a probabilistic framework inspired by Kalman filtering, applying filters to remove uncertain initial estimates, and integrating bidirectional flow with long-term matches.

EgoPoints[7] introduces the first large-scale benchmark for point tracking in egocentric videos, annotating 4.7k challenging tracks across 517 sequences with 9x more out-of-view points and 59x more re-identification points compared to TAP-Vid DAVIS. When third-person-trained methods were evaluated in EgoPoints [7], the performance dropped significantly, indicating there exist a domain gap between third-personn videos and the egocentric videos.

Our work is different from the previous methods. All the previous works[3],[6],[2] focuses on third person videos, but we focus on egocentric videos. We try to access how the foundation models DINOv2, CroCov2 out the box perform on point tracking task in the egocentric video setting. We observe that merely fine-tuning the foundation models on egocentric data does not boost their performance, and hence the naive adaptation does not close the domain gap.

3 Method

We utilize frozen DINOv2 or CroCov2 backbones to extract patch-level features from image pairs and then learn an upsampling function, comprising a convolutional layer followed by bilinear interpolation, to project these features into dense, per-pixel representations at the resolution of the original images. Additionally, we introduce a delta network that learns a residual offset between the patch features of the two images. This network leverages cross-attention between the token embeddings and convolutional features to infer displacement vectors that align keypoint features from the second image with their counterparts in the first. The delta output is upsampled into a dense feature map and added as a residual to the second image's dense features to model the motion between corresponding keypoint features. We then optimized the model using either L2 or cosine loss to learn to predict accurate feature displacements that align keypoint features across image pairs.

To leverage the model for tracking, we first pass the image pair through the network to obtain aligned dense feature maps for both images. For each keypoint in the first image, we bilinearly sample its corresponding feature from the first image's feature map. We then search for the most similar feature in the second image's feature map and use the location of this best match to determine where the keypoint has moved, effectively tracking its displacement across frames. Figure 1 shows the architecture of our model.

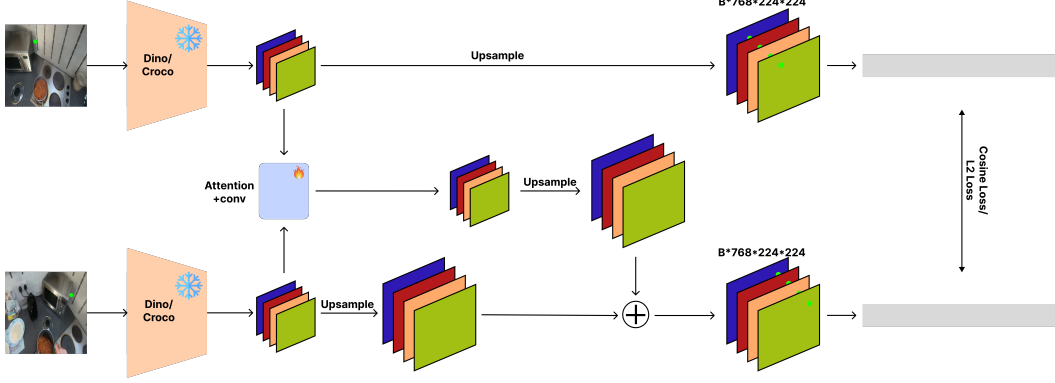


Figure 1: Foundation model adaptation with our Delta Net

4 Experiments

In this section, we detail our experimental setup and present the results of our experiments, focusing on both the fine-tuning of foundation models—CroCoV2 [8] and DINOv2 [9]—and the utilization of their pre-trained versions for point tracking in egocentric videos.

Datasets: Our models were trained using the training split of the EgoPoints [7] dataset. To create training pairs, we applied a windowing strategy that samples video frames at intervals of 8 (i.e., selecting frames at multiples of 8), resulting in temporally spaced image pairs. All image pairs were resized to 224×224 resolution to match the input requirements of DINOv2 [9] and CroCoV2 [8]. The corresponding 2D keypoints were also rescaled accordingly to remain consistent with the resized images.

Baselines: Our baselines consist of vanilla pre-trained models without any fine-tuning. We used the final token features from the last transformer layer of the encoder and up-sampled them to the input image resolution using bilinear interpolation. For point tracking, we extract features at keypoint locations in the current frame and match them with their nearest neighbors in the feature volume of the next frame.

Evaluation Metrics: Our evaluation metric is **Accuracy at different pixel thresholds**, τ . This metric evaluates how accurately keypoints are tracked by measuring the percentage of predicted points that lie within a certain Euclidean distance from the ground truth. We report accuracy at thresholds of 8px, 16px, and 24px on a subset of the EgoPoints [7] validation split.

Formally, let $\hat{\mathbf{p}}_i$ denote the predicted 2D location of keypoint i , and $\mathbf{p}_i$ the corresponding ground-truth location. Then, the accuracy at threshold τ is defined as:

$$Accuracy_{\tau} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\|\hat{\mathbf{p}}_i - \mathbf{p}_i\|_2 \leq \tau]$$

where N is the total number of keypoints, and $\mathbb{1}[\cdot]$ is the indicator function that equals 1 if the condition inside is true and 0 otherwise. This metric allows us to quantify the precision of keypoint tracking at various levels of tolerance.

Training: We trained our models using the AdamW optimizer (with default betas $\beta_1 = 0.9$, $\beta_2 = 0.999$) and a learning rate of 1×10^{-5} , with a batch size of 32. Training was carried out over 100 epochs on one A100 GPU for six hours. The DINOv2 and CroCoV2 backbones were initialized with pre-trained weights, and only the parameters of the delta network were updated during training. We applied color jittering as the data augmentation. The models are trained using either L2 loss or cosine similarity loss. We fine-tuned CroCoV2 [8], by optimizing the last transformer layer in the encoder on EgoPoints training data using the original Self Supervised Learning (SSL) loss.

4.1 Results

Our results, summarized in Table 4.1, highlight the impact of delta modeling and loss choice on point tracking accuracy. For both CroCov2 and DINOv2 backbones, the pre-trained models achieve strong baseline performance, with DINOv2 (pre-trained) achieving the highest overall accuracy (avg 34.54). Among fine-tuned models, CroCov2 with cosine loss performs best (avg 33.39), showing a significant improvement over L2 and SSL losses. In contrast, DINOv2 exhibits lower performance when fine-tuned. Surprisingly, our fine-tuned DINOv2 variants consistently underperform compared to the pre-trained baseline. This highlights a fundamental mismatch between DINOv2’s training objective, global image-level representation learning, and the fine-grained, spatially sensitive demands of point tracking. Incorporating the delta network yields mixed results, indicating that the benefit depends on the loss function and backbone architecture. As shown in prior work [6], learning a delta to model the displacement in feature space between keypoint features of image pairs can degrade the rich representations learned by the original DINOv2 features. Previous approaches [6] addressed this by introducing a feature preservation loss, which constrains the learned delta to prevent the features from drifting too far from the pre-trained DINOv2 representations. We believe this is a promising direction and plan to explore it further in future work.

Another reason for the poor performance of pre-trained foundation models on egocentric videos lies in their inherent lack of motion understanding. Models like DINOv2 and CroCov2 are trained primarily on static image collections, which means that they are not exposed to the temporal dynamics and motion cues that are crucial in video understanding, particularly in egocentric settings. In such scenarios, where the camera undergoes rapid and unpredictable movement and objects frequently appear, disappear, or change scale, a deep understanding of motion is essential for consistent point tracking. Without temporal supervision or motion-centric pretraining, these models struggle to learn how features evolve over time, leading to poor performance in maintaining reliable correspondences. This missing capacity to model egomotion, occlusions, and temporal consistency highlights a fundamental gap in current foundation models when applied to dynamic video contexts.

We also performed a qualitative evaluation to gain deeper insight into the behavior of the models. Figures 3, 4, and 5 show the point tracking results of various models on a representative sequence from the EgoPoints [7] dataset. As illustrated, all models struggle to reliably track dynamic objects across frames. For comparison, the corresponding ground truth tracks are shown in Figure 2, with dotted lines indicating occluded points. These observations support our hypothesis that explicitly modeling motion is essential when adapting foundation models for point tracking in egocentric videos.

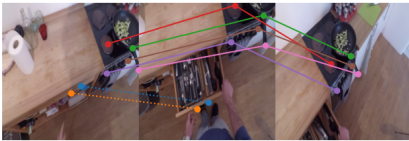


Figure 2: Ground truth tracks



Figure 3: DINOv2 (Pre-trained) tracks



Figure 4: CroCov2 (Pre-trained) tracks

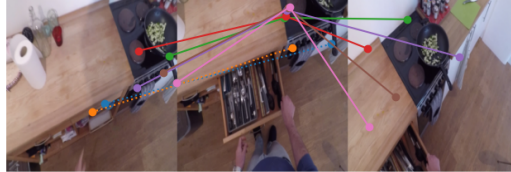


Figure 5: CroCov2 (Cosine Loss) fine-tuned tracks

Model	Loss	Delta	acc@8px (↑)	acc@16px (↑)	acc@24px (↑)	avg (↑)
CroCov2	L2	✓	12.77	24.09	32.33	23.06
	L2	✗	13.89	25.28	34.13	24.44
	Cosine	✗	20.96	35.66	43.54	33.39
	SSL loss	✗	12.46	24.35	31.78	22.87
CroCov2 (Pre-trained)	–	✗	20.10	34.31	42.13	32.18
DINOv2	L2	✓	1.58	6.24	10.41	6.08
	L2	✗	3.78	10.74	15.12	9.09
	Cosine	✗	4.15	12.17	17.69	11.34
DINOv2 (Pre-trained)	–	✗	19.57	37.04	47.00	34.54

Table 1: Accuracy at different pixel thresholds **Red**: Best , **Blue**: Second best

5 Conclusion

In this work, we investigated the point tracking capabilities of pre-trained and fine-tuned foundation models, specifically DINOv2 and CroCov2, on egocentric video data. Our results indicate that while pre-trained models demonstrate promising baseline performance, fine-tuning them for point tracking can lead to performance degradation, particularly for models like DINOv2 that were not originally optimized for fine-grained spatial tasks. We identified key challenges, including the lack of motion understanding and the risk of drifting from pretrained feature spaces when learning displacement deltas, which limit these models’ effectiveness in dynamic egocentric scenarios. Looking forward, we believe future research should focus on integrating temporal supervision and motion cues into foundation model training. One promising direction is to leverage 3D-aware foundation models such as DUST3R [10], MAST3R [11], and VGGT [12], which are better equipped to capture depth and scene geometry. Incorporating optical flow as an additional supervisory signal may further enhance motion understanding. By combining spatial, temporal, and geometric cues, future models may achieve more robust and consistent point tracking in the complex, dynamic environments typical of egocentric videos.

References

- [1] Adam W. Harley, Zhaoyuan Fang, and Katerina Fragkiadaki. Particle video revisited: Tracking through occlusions using point trajectories, 2022.
- [2] Nikita Karaev, Iurii Makarov, Jianyuan Wang, Natalia Neverova, Andrea Vedaldi, and Christian Rupprecht. Cotracker3: Simpler and better point tracking by pseudo-labelling real videos, 2024.

- [3] Tingyang Zhang, Chen Wang, Zhiyang Dou, Qingzhe Gao, Jiahui Lei, Baoquan Chen, and Lingjie Liu. Protracker: Probabilistic integration for robust and accurate point tracking, 2025.
- [4] Carl Doersch, Ankush Gupta, Larisa Markeeva, Adrià Recasens, Lucas Smaira, Yusuf Aytar, João Carreira, Andrew Zisserman, and Yi Yang. Tap-vid: A benchmark for tracking any point in a video, 2023.
- [5] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset, 2018.
- [6] Narek Tumanyan, Assaf Singer, Shai Bagon, and Tali Dekel. Dino-tracker: Taming dino for self-supervised point tracking in a single video, 2024.
- [7] Ahmad Darkhalil, Rhodri Guerrier, Adam W Harley, and Dima Damen. Ego-points: Advancing point tracking for egocentric videos. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 8556–8565. IEEE, 2025.
- [8] Philippe Weinzaepfel, Thomas Lucas, Vincent Leroy, Yohann Cabon, Vaibhav Arora, Romain Brégier, Gabriela Csurka, Leonid Antsfeld, Boris Chidlovskii, and Jérôme Revaud. Croco v2: Improved cross-view completion pre-training for stereo matching and optical flow. *arXiv preprint arXiv:2211.10408*, 2022.
- [9] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- [10] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.
- [11] Vincent Leroy, Yohann Cabon, and Jérôme Revaud. Grounding image matching in 3d with mast3r. In *European Conference on Computer Vision*, pages 71–91. Springer, 2024.
- [12] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. *arXiv preprint arXiv:2503.11651*, 2025.